Koopman Autoencoders Reveal Interpretable Eigenmodes in LLM Latent Dynamics

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Abstract

Autoregressive large language models (LLMs) have been extensively studied through static representational analyses, yet the dynamics of their token-wise activation trajectories remain insufficiently characterized. Extending from Latent Spectral Dynamics that establishes a stochastic autonomous dynamical systems framework on windowed residual activations, we refine the data-constrained and less-interpretable Hankel-EDMD method with Koopman Autoencoders (KAEs) – deep encoder–decoder networks that learn nonlinear observable bases and a finite-dimensional linear Koopman operator simultaneously. Applied to delay-windowed residual stream trajectories from multiple models across mathematical reasoning and narrative generation, KAEs yield three principal findings: (1) an *invariant eigenspectrum* at fixed latent dimension that persists across datasets and model families, suggesting a universal autoregressive dynamical backbone; (2) a *resolution scaling* phenomenon in which increasing the encoder latent dimension progressively reveals finer transient structure while preserving previously recovered modes; and (3) an *interpretability pipeline*—inspired by sparse autoencoder feature analysis—that taxonomizes learned Koopman eigenmodes into a structural–transient hierarchy. High-energy backbone and oscillatory modes track formatting boundaries and problem–answer transitions, while low-energy transient modes capture brief and task-specific computational events such as equation-to-result pivots. Probing confirms that the Koopman latent space can be linearly separable for the reasoning phase, arithmetic context, and trajectory correctness, while partial causal ablation shows that mode removal produces localized perturbations in the residual stream. These results establish that autoregressive LLM generation admits a spectral decomposition into a dominant structural backbone and sparse reasoning-specific transients, strengthening a dynamical-systems foundation for interpretability.

1 Introduction

Autoregressive large language models (LLMs) are increasingly deployed as autonomous agents capable of advanced reasoning, yet their internal computational processes remain opaque. Mechanistic interpretability has produced powerful tools for dissecting Transformer internals—circuit analysis [7], probing methods [4], representation engineering [24], and sparse autoencoders [17]—but these approaches operate on *static* snapshots: fixed layers, token positions, or weight states. Characteristics of the *temporal evolution* of activations across the token-by-token generation process, such as how the residual stream transitions between computational phases, oscillates during uncertainty, or converges toward an answer, are invisible to methods that analyze individual snapshots in isolation.

A recent line of work has begun to address this gap by framing neural computation through a dynamical-systems lens. Deep Equilibrium Models [3] and Neural ODEs [6] established that neural

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computation can be analyzed as a dynamical system. The Latent Spectral Dynamics (LSD) framework [21] formalized autoregressive LLM generation as a stochastic, autonomous, discrete-time dynamical system on time-delay-embedded residual stream states, and applied Koopman operator theory via Hankel-Extended Dynamic Mode Decomposition (Hankel-EDMD) to extract spectral signatures of truthful versus untruthful generation. That work demonstrated that truthful trajectories converge toward a stable invariant measure with near-unity real eigenvalues, while untruthful trajectories exhibit oscillatory dynamics driven by complex-conjugate eigenvalue pairs—a distinction invisible to static representational analyses.

However, the Hankel-EDMD approach faces critical scaling limitations. Its reliance on exact matrix factorizations over massive, uncompressed coordinate bases—with Gram matrices of dimension $N \times N$ and ambient observable dimensions $D = Wd$ exceeding 25,000—makes it prohibitively expensive when scaling to larger models or longer generation contexts. Furthermore, the coordinate basis is restricted to linear observables, potentially missing nonlinear dynamical structure in the residual stream.

In this work, we address both limitations by replacing Hankel-EDMD with Koopman Autoencoders (KAEs): deep encoder–decoder networks that jointly learn a nonlinear observable basis and a finite-dimensional linear Koopman operator. This substitution offers four key advantages: (i) *computational and data scalability*, achieved by bypassing cubic-cost matrix factorization in favor of stochastic gradient descent; (ii) the *capacity to learn nonlinear observables* that capture dynamics invisible to linear bases; (iii) a *tolerance for small delay windows*, as the learned encoder extracts richer features from each activation vector; and (iv) *enhanced interpretability*, which allows one to study Koopman eigenmodes via probing and ablation tests by leveraging the decoder's reconstruction of the original state.

We apply this framework to residual stream trajectories from three 14B-parameter LLMs across mathematical reasoning (GSM8K, MATH500) and open-text generation (WikiText-103), and develop a systematic *interpretability pipeline* for the learned Koopman eigenmodes.

Our main contributions are:

1. **Scalable Operator Learning.** We introduce Koopman Autoencoders (KAEs) as a deep learning alternative to Hankel-EDMD for approximating the Koopman operator on LLM residual stream trajectories, bypassing the cubic-cost matrix factorization that limited the prior framework to small sample regimes (Section 3).

2. **Interpretability Pipeline.** Adapting the methodology of sparse autoencoder (SAE) feature analysis to the dynamical setting, we design a six-step pipeline for interpreting learned Koopman eigenmodes: eigenmode taxonomy, max-activating segment extraction, LLM-assisted feature labeling, linear probing validation, and causal ablation (Section 5.4).

3. **Invariant Eigenspectra.** KAEs trained independently on different datasets (GSM8K, MATH500, WikiText-103) and model families (Qwen2.5, DeepSeek-R1-Distill, Mistral-3) recover similar spectral geometries at fixed latent dimension, suggesting a universal dynamical backbone of autoregressive generation (Section 5.2).

4. **Resolution Scaling.** Increasing the latent dimension from $r = 64$ to $r = 512$ progressively reveals finer transient structure while preserving previously recovered modes, establishing a spectral hierarchy from universal to task-specific dynamics (Section 5.3).

5. **Structural–Transient Mode Hierarchy.** The interpretability pipeline reveals that high-energy backbone and oscillatory modes track the structural skeleton of generation—problem boundaries, formatting templates, turn transitions—while low-energy transient modes capture brief, reasoning-specific computational events such as equation-to-result pivots (Section 5.4).

2 Background: Dynamical Systems Formulation

We briefly review the dynamical formulation of Anonymous [21], which provides the mathematical foundation for our extension.

2.1 Markovian Dynamics and Phase Space Reconstruction

At generation step k, an autoregressive Transformer maintains a KV cache S_k that constitutes an exact Markov state: $P_\phi(x_{k+1}|x_{1:k}) = P_\phi(x_{k+1}|S_k)$, where ϕ is the fixed parameter of the model. Because S_k is impractically high-dimensional for operator estimation, we construct a fixed-dimensional surrogate via Hankel time-delay embedding of the final-layer residual stream $h_k \in \mathbb{R}^d$:

$$H_k = [h_{k-W+1}^\top, \ldots, h_k^\top]^\top \in \mathbb{R}^{Wd} \tag{1}$$

Under the measure-theoretic phase-space reconstruction theorem of Botvinick-Greenhouse et al. [5], for generic observation maps and sufficient window size W, the delay-embedded state H_k preserves the statistical structure of S_k.

2.2 Autonomous Dynamics and the Koopman Operator

The original LSD framework formulates a stochastic autonomous system via Gumbel-Max reparameterization, $H_{k+1} = F_{\phi,\theta^*}(H_k, \Xi_k)$, where Ξ_k is exogenous Gumbel noise and θ^* indexes the prompting regime. Under greedy decoding ($\tau = 0$), which all experiments in this paper adopt, the noise vanishes and the system becomes purely deterministic:

$$H_{k+1} = F_{\phi,\theta^*}(H_k) \tag{2}$$

An *observable* is any scalar measurement of the state, $g : \mathbb{R}^{Wd} \to \mathbb{C}$. The *Koopman operator* $\mathcal{K}_{\theta^*}$ advances observables forward by one timestep, lifting the nonlinear state dynamics into a linear evolution on functions:

$$(\mathcal{K}_{\theta^*} g)(H_k) = g(F_{\phi,\theta^*}(H_k)) \tag{3}$$

This deterministic setting ensures that all spectral structure recovered from the learned operator reflects the model's computation rather than sampling noise. Notably, LSD approximated $\mathcal{K}_{\theta^*}$ via Hankel-EDMD with Tikhonov regularization.

3 Method: Koopman Autoencoders for LLM Dynamics

3.1 Architecture

We replace Hankel-EDMD with a Koopman Autoencoder (KAE), which consists of three components:

- An **encoder** $\phi : \mathbb{R}^{Wd} \to \mathbb{R}^r$ serving as the learned observable dictionary,
- A **learnable matrix** $K \in \mathbb{R}^{r\times r}$ approximating the Koopman operator in the latent space,
- A **decoder** $\psi : \mathbb{R}^r \to \mathbb{R}^{Wd}$ reconstructing Hankel states from latent representations.

The system is trained by minimizing a combined objective:

$$\mathcal{L} = \underbrace{\|H_k - \psi(\phi(H_k))\|^2}_{\text{Reconstruction}} + \alpha \underbrace{\|\phi(H_{k+1}) - K\phi(H_k)\|^2}_{\text{Linear dynamics}} + \beta \underbrace{\sum_{n=1}^{N} \|\phi(H_{k+n}) - K^n\phi(H_k)\|^2}_{\text{Multi-step rollout}} \tag{4}$$

where α and β are tunable parameters. Specifically, the reconstruction term prevents degenerate encodings, the linear dynamics term enforces that transitions are linear in the learned latent space, and the rollout term stabilizes eigenvalues over longer horizons.

3.2 Architecture

We study two KAE variants operating directly on the full Hankel state $H_k \in \mathbb{R}^{Wd}$ without dimensionality reduction. A **linear KAE** uses single-layer affine encoder and decoder ($W_{\text{enc}} \in \mathbb{R}^{r\times Wd}$, $W_{\text{dec}} \in \mathbb{R}^{Wd\times r}$), serving as a learned linear compression baseline. A **nonlinear KAE** uses a two-layer MLP encoder ($\mathbb{R}^{Wd} \to \mathbb{R}^{128}$ with ReLU $\to \mathbb{R}^r$) and a symmetric decoder. For $W = 8$ and $d = 5120$, the input dimension is $Wd = 40{,}960$; the nonlinear variant has $\sim$10.5M parameters at $r = 64$ and $\sim$10.6M at $r = 256$, dominated by the first encoder layer. The Koopman matrix $K \in \mathbb{R}^{r\times r}$ adds r^2 parameters. In preliminary experiments (Section **??**), the nonlinear KAE consistently outperformed the linear variant in rollout MSE, confirming that nonlinear observables capture dynamical structure beyond linear compression; all subsequent results use the nonlinear architecture.

3.3 Spectral Analysis

The learned matrix K admits eigendecomposition $K = V\Lambda V^{-1}$ whenever it is diagonalizable. Since a general $K \in \mathbb{R}^{r\times r}$ is not guaranteed to be diagonalizable, we verify this empirically: if the condition number of the eigenvector matrix $\kappa(V)$ exceeds 10^3—indicating near-defectiveness—we fall back to the Schur decomposition $K = QTQ^*$, where T is upper triangular and Q is unitary, which exists for any square matrix and yields the same eigenvalues on the diagonal of T. In all experiments reported here, the learned operators satisfy $\kappa(V) \in [194, 328]$, so standard eigendecomposition is used throughout.

We examine the following components during spectral analysis of K:

- **Eigenvalues** $\lambda_j \in \mathbb{C}$ characterize the temporal behavior of each mode: $|\lambda_j|$ governs its decay rate per token step, and $\arg(\lambda_j)$ its oscillation frequency (period $= 2\pi/|\arg(\lambda_j)|$ tokens).
- **Modal coefficients** $c_j(k) = (V^{-1}\phi(H_k))_j \in \mathbb{C}$ measure the activation of mode j at token k. Each c_j is a scalar time series that evolves as $c_j(k) \approx \lambda_j^k\, c_j(0)$ under the learned dynamics, corresponding to the finite-dimensional approximation of the j-th Koopman eigenfunction evaluated along the trajectory.
- **Koopman modes** $\tilde{v}_j = J_\psi(\bar{z})\, v_j \in \mathbb{R}^{Wd}$, obtained by mapping latent eigenvectors through the decoder Jacobian, identify which residual stream dimensions and token positions participate in each temporal mode.

The eigenvalues carry the temporal structure (frequency, decay rate) and the modes carry the spatial structure (which dimensions participate), grounding the interpretability analysis in Section 5.4.

4 Experimental Setup

Models. We use three 14B-parameter instruction-tuned LLMs: Qwen2.5-14B-Instruct, DeepSeek-R1-Distill-Qwen-14B, and Mistral-3-14B.

Datasets. For each model, we extract final-layer residual stream trajectories ($d = 5120$) from 500 sequences in each of three domains: (i) **GSM8K**, grade-school math word problems requiring multi-step arithmetic; (ii) **MATH500**, competition-level mathematics with heavier LaTeX formatting; and (iii) **WikiText-103**, open-text continuation serving as a non-reasoning control. All sequences are capped at 512 tokens under greedy decoding ($\tau = 0$).

KAE Configuration. We train a nonlinear KAE with Hankel window $W = 8$ and sweep the latent dimension across $r \in \{64, 128, 256, 512\}$, using a multi-step rollout horizon of $N = 8$. Training uses AdamW with learning rate 10^{-4} and batch size 256, with an 80/10/10 train/validation/test split stratified by trajectory. We evaluate fit quality via the relative multi-step rollout MSE:

$$\text{Rel-MSE}(n) = \frac{\sum_i \|\phi(H_{k_i+n}) - K^n\phi(H_{k_i})\|^2}{\sum_i \|\phi(H_{k_i+n})\|^2} \tag{5}$$

which measures the fraction of transition variance unexplained by the learned operator at horizon n, computed on held-out test trajectories.

5 Results

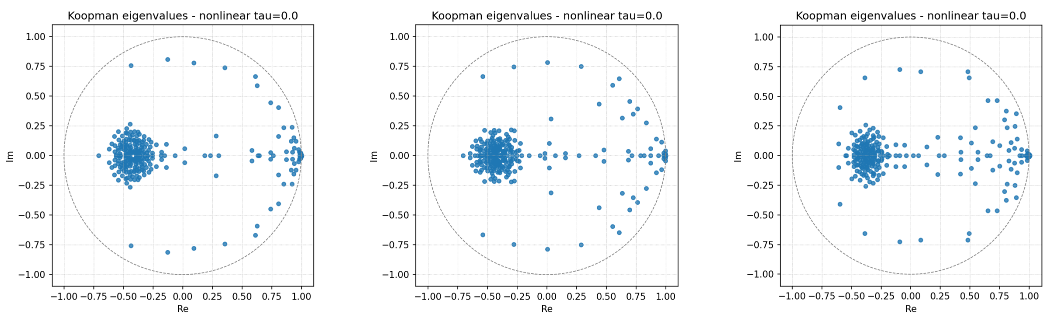
5.1 Dynamical Fit Quality

We first verify that the KAE successfully learns a Koopman operator governing the residual stream dynamics. Table 1 reports rollout MSE across rollout horizons for Qwen2.5-14B on GSM8K.

The stable degradation from 0.481 (1-step) to 0.727 (8-step) confirms that the learned operator captures the dominant Markovian transition structure, consistent with the original LSD finding that coordinate-basis EDMD operators support stable multi-step rollouts.

Table 1: KAE rollout MSE on GSM8K (Qwen2.5-14B-Instruct, $r = 256$, $W = 8$).

Horizon	**Train MSE**	**Val MSE**	**Test MSE**
1-step	0.406	0.448	0.478
4-step	0.418	0.512	0.546
8-step	0.476	0.686	0.727



(a) GSM8K (b) MATH500 (c) WikiText-103

Figure 1: Koopman eigenspectra at $r = 256$, $W = 8$ across three datasets (Qwen2.5-14B-Instruct). All three spectra share the same three-component geometry: a near-unity real backbone cluster (right), a dense negative-real cluster near $\mathrm{Re} \approx -0.4$ corresponding to period-2 oscillations (left), and scattered complex-conjugate pairs throughout the unit disk. The dashed circle marks $|\lambda| = 1$.

Comparison to EDMD baseline. On the truthful/untruthful regime from Anonymous [21], the nonlinear KAE achieves 1-step test MSE of 0.525 (untruthful) and 0.636 (truthful), outperforming the linear KAE variant (0.685 and 0.789 respectively), confirming that the nonlinear encoder captures dynamical structure beyond linear compression. However, the KAE's spectral radius slightly exceeds unity ($\rho \approx 1.02$–1.04), indicating marginal global instability that does not affect short-horizon analysis but warrants spectral regularization in future work.

5.2 Invariant Eigenspectra Across Datasets

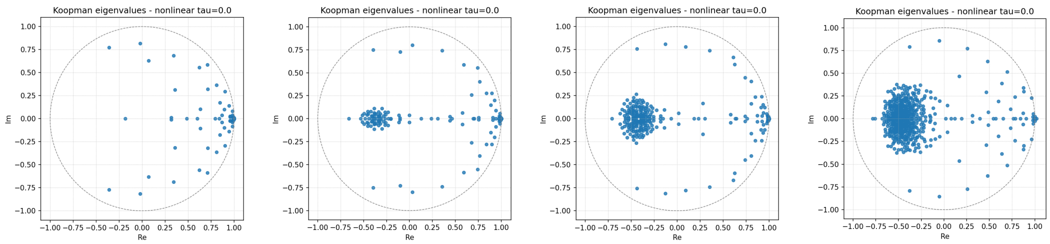
At fixed latent dimension $r = 256$, KAEs trained separately on GSM8K, MATH500, and WikiText-103 from Qwen2.5-14B yield strikingly similar spectral geometries (Figure 1). All three spectra share: (i) a dense near-unity real cluster corresponding to slowly evolving backbone modes; (ii) a negative-real cluster near Re ≈ -0.4 corresponding to period-2 oscillations; and (iii) scattered complex-conjugate pairs throughout the unit disk.

Because each KAE learns its own nonlinear observable basis, these operators inhabit different latent spaces, making direct numerical comparison ill-posed. The spectral similarity is therefore a qualitative observation; quantitative confirmation would require coordinate-free invariants such as the Wasserstein distance between eigenvalue distributions or cross-prediction experiments with a shared encoder. Nevertheless, the persistence of the eigenspectrum across a math reasoning task, a competition math task, and an open-text continuation task—domains with very different token-level statistics—is suggestive of a universal autoregressive dynamical backbone.

5.3 Resolution Scaling of Eigenspectra

We hypothesize that the KAE captures the highest-variance dynamical components first, with eigenvalues at low r reflecting universal autoregressive structure and higher r revealing progressively finer, potentially task-specific oscillatory modes.

Figure 2 shows the operator eigenspectra of KAEs trained on GSM8K at $r \in \{64, 128, 256, 512\}$. At $r = 64$, eigenvalues are sparse with the near-unity backbone already present but few complex modes. As r increases, a dense cluster progressively forms around Re ≈ -0.4, emerging at $r = 128$ and growing substantially through $r = 512$. The near-unity real backbone persists stably across all



(a) $r = 64$ (b) $r = 128$ (c) $r = 256$ (d) $r = 512$

Figure 2: Resolution scaling of the Koopman eigenspectrum (Qwen2.5-14B-Instruct, GSM8K, $W = 8$). At $r = 64$, eigenvalues are sparse with the near-unity backbone already present. As r increases, a dense cluster forms progressively around $\mathrm{Re} \approx -0.4$, while the backbone persists stably across all resolutions. The dashed circle marks $|\lambda| = 1$.

resolutions. The accumulation of period-2 oscillatory modes is the primary structural change with increasing r.

This resolution scaling replicates across all three models tested (Qwen2.5, DeepSeek-R1-Distill, Mistral-3), strengthening the claim that it reflects an intrinsic property of autoregressive generation rather than a model-specific artifact.

5.4 Interpreting the Structural–Transient Mode Hierarchy

We develop a six-step interpretability pipeline, drawing methodological parallels to SAE feature analysis, to characterize the learned Koopman eigenmodes.

5.4.1 Step 1: Eigenmode Taxonomy

For each mode j, we record the eigenvalue magnitude $|\lambda_j|$, phase $\theta_j = \arg(\lambda_j)$, period $2\pi/|\theta_j|$ when $\theta_j \neq 0$, and an empirical energy fraction E_j. The unnormalized energy is computed by taking, for each trajectory, the variance over Hankel positions of $|c_j(k)|^2$, averaging this variance across trajectories, and normalizing across modes so that $\sum_j E_j = 1$.

We use a simple rule-based taxonomy as a coarse descriptive summary of the spectrum. Let $q_{0.05}$ and $q_{0.50}$ denote the 5th percentile and median of the mode energy fractions $\{E_j\}$. Modes are assigned in the following order:

- **Transient modes**: modes with $|\lambda_j| < 0.5$ or $E_j \leq q_{0.05}$. These modes either decay rapidly or have negligible empirical activation energy.
- **Backbone modes**: remaining modes with $|\lambda_j| \geq 0.9$, near-real phase $|\theta_j| \leq 0.05$, and $E_j \geq q_{0.50}$. These modes are high-energy, slowly decaying directions close to the positive real axis.
- **Oscillatory modes**: all remaining modes. This category contains the complex and intermediate-magnitude modes, including conjugate pairs when present, and serves as a coarse bin for temporal structure not captured by the backbone or transient criteria.

This taxonomy is intended as a first-pass spectral summary; finer classification is left to future work.

Table 2: Eigenmode taxonomy at $r = 256$, $W = 8$ across datasets (Qwen2.5-14B-Instruct).

Dataset	$\rho(K)$	$\kappa(V)$	Backbone	Oscillatory	Transient
GSM8K	1.003	194.5	23	73	160
MATH500	1.002	277.2	16	72	168
WikiText-103	1.006	327.5	35	58	163

Table 2 summarizes the mode distribution across datasets at $r = 256$. The three-tier hierarchy is consistent across all datasets. WikiText-103 has the most backbone modes (35 vs. 16–23), reflecting greater document-level structural diversity compared to the relatively uniform problem–solution templates of math datasets.

5.4.2 Step 2: Max-Activating Segments

For each mode j, we identify text spans where the modal coefficient $|c_j(k)|$ is unusually large. Specifically, we threshold activations at $\tau_j = \mu_j + 1.5\sigma_j$, extract contiguous runs of at least five Hankel positions above threshold, and retain the top 10 text-aligned segments per mode ranked by peak activation.

On GSM8K, this procedure scans all 256 modes and yields 2560 stored segments. The extracted windows concentrate around recurring procedural events in generation: problem-to-solution onset, variable or equation setup, arithmetic execution, final-answer emission, follow-up-question handoff, and prompt reset. For example, mode 90 activates near follow-up-question boundaries, modes 32/33 near final-answer production and transition to the next prompt, and modes 47/48 and 51/52 near problem-boundary or solution-onset events. Detailed extraction statistics and representative examples are reported in Appendix E.

5.4.3 Step 3: Feature Labeling

We label a high-priority panel of 25 modes selected by energy. For each mode, the annotator receives its spectral metadata together with the top six max-activating segments from Step 2, yielding 150 mode-context examples in total.

The resulting GSM8K labels are dominated by procedural and mixed features: 13 procedural labels, 10 mixed labels, and 2 semantic labels, with 13 high-confidence and 12 medium-confidence annotations. Common labels include answer finalization, problem-statement-to-solution transition, follow-up question parsing, variable/equation setup, and turn-boundary detection. This pattern suggests that KAE eigenmodes primarily organize generation dynamics rather than forming a static semantic dictionary. The few semantic labels appear narrowly tied to repeated surface templates, so we treat Step 3 labels as structured hypotheses that require probing and ablation for validation. Additional labeling details are given in Appendix E.

5.4.4 Step 4: Probing Validation

We train linear probes on two representations of the KAE state: the raw latent coordinates $\phi(H_k)$ and the eigenbasis coordinates $c(k)$, represented by their real and imaginary parts. Rather than relying on heuristic trajectory-derived targets (reasoning phase, arithmetic context, correctness, distance-to-answer), which we retain only as diagnostics in Appendix C, we focus validation on binary targets derived from the Step 3 max-activation labels for five representative eigenmodes. For each mode, a token is labeled positive if it falls within that mode's top max-activating window; classification is reported as balanced accuracy (BA) against a 0.500 chance baseline. Table 3 reports results for the transient boundary mode 90, the answer-finalization pair 32/33, the problem-to-solution boundary pairs 47/48 and 51/52, and the semantic control pair 10/11.

Table 3: Label-derived linear probing results on GSM8K (Qwen2.5-14B-Instruct, $r = 256$). All targets are binary (chance = 0.500).

Mode(s)	Label-Derived Target	Raw BA	Eigen BA	Positives
90	Follow-up-question boundary	0.977	0.977	298
32/33	Answer finalization / prompt transition	0.772	0.753	134
47/48	Problem-to-solution boundary	0.977	0.974	417
51/52	Question-boundary / solution onset	0.997	0.997	369
10/11	Surface template ("be denoted")	0.999	0.999	10

The boundary and solution-onset modes (90, 47/48, 51/52) are linearly recoverable with BA > 0.97, confirming that the generation events identified in Step 3 are accessible in the KAE latent space. The answer-finalization pair 32/33 is recoverable but weaker (BA ≈ 0.77), partly because the positive class is sparser. The control pair 10/11 reaches near-perfect BA on only 10 positives, reinforcing that high probe accuracy can reflect a narrow surface-template event rather than broad semantic structure.

These probes serve as consistency and selectivity checks for the eigenmode labels, not as fully independent validation. The value of the eigenbasis is not higher aggregate probe accuracy—raw and eigenbasis scores are nearly identical throughout—but temporal factorization: individual modes expose recurring dynamical patterns that can be inspected, labeled, and targeted by downstream intervention.

5.4.5 Step 5: Causal Ablation (Partial)

To move beyond correlational probing, we propose a three-tier causal intervention framework to establish the functional role of the discovered eigenmodes. *Tier 1* identifies dynamically necessary modes by zeroing out an eigenmode ($c'_j = 0$), rolling out the reconstructed latent state, and measuring the increase in multi-step rollout MSE. *Tier 2* decodes the ablated state to compute the perturbation ΔH, mapping the effect to specific residual stream dimensions and token positions. *Tier 3* verifies causal relevance by injecting this residual perturbation into the LLM prior to unembedding and observing the resulting logit shifts, calibrated against a random perturbation of equal norm.

Due to computational constraints, we present partial results focusing on Tier 2. Tier 1 results are deferred because the KAE incorrect execution of pipeline produced NaN baselines. Tier 3 is deferred because injecting logit shifts dynamically requires holding both the full 14B LLM and the KAE in memory simultaneously across multiple perturbation scales; we plan to address this in future work via offline pre-computation of the perturbations.

We conducted Tier 2 ablation on the complex-conjugate pair modes 32/33, which were labeled in Step 3 as answer-finalization detectors on GSM8K (activating near the "####" delimiter and numeric answer). Zeroing these modes and mapping the perturbation back to the 5120-dimensional residual stream yields a substantial mean L_2 norm of 210.44 (with a max L_2 of 845.12). Notably, the perturbation is highly localized: the top five affected dimensions account for a disproportionate share of the overall effect. The most heavily perturbed coordinates are dimension 2862 (mean shift $+59.87$), dimension 1372 ($+28.80$), dimension 1661 (-24.42), dimension 2442 ($+20.09$), and dimension 3183 ($+18.75$). This high localization indicates that modes 32/33 do not diffusely perturb the internal representation, but instead target a specific residual-stream subspace—likely corresponding to answer-token logits or formatting-control features—which remains to be causally confirmed via Tier 3 injection.

5.5 Cross-Architecture Consistency

All three findings—invariant eigenspectra, resolution scaling, and the structural–transient hierarchy—replicate across Qwen2.5-14B-Instruct, DeepSeek-R1-Distill-Qwen-14B, and Mistral-3-14B. The near-unity real backbone, negative-real cluster, and accumulation of complex-conjugate pairs with increasing r appear in each model's eigenspectra with only minor quantitative variation. This cross-architecture consistency strengthens the claim that these spectral structures reflect intrinsic properties of autoregressive generation.

6 Related Work

Dynamical Systems Perspectives on Neural Networks. Deep Equilibrium Models [3] and Neural ODEs [6] established that neural computation can be analyzed as a dynamical system. In-context learning has been framed as implicit Bayesian inference [18], and finite-state Markov chain models of bounded-context LLMs have been developed [20, 8]. The Latent Spectral Dynamics framework [21] extracts Koopman operators directly from autoregressive LLM activation trajectories, demonstrating that spectral signatures invisible in the output token stream can be recovered from the residual stream.

Chain-of-Thought Monitoring and Faithfulness. A growing literature monitors LLM reasoning by perturbing surface-level chain-of-thought traces and measuring downstream effects [25–27], revealing that CoT is frequently unfaithful to the model's internal computation. Complementary work probes *latent* representations directly: Feng et al. [28] extract propositional world states from activations, finding that internal representations remain faithful even when surface outputs are not. Our approach shares this emphasis on latent monitoring but extracts the *continuous temporal dynamics* governing how representations evolve across the full generation trajectory, rather than decoding discrete states at individual token positions.

Spectral and Frequency-Domain Analyses. HSAD [11] applies FFT across Transformer layers to detect hallucination-correlated spectral anomalies; Akrout [1] apply DMD to sentence-level embeddings of hallucinated paragraphs. Geometry of Reason [15] analyzes graph-Laplacian structure in attention, and NerVE [9] tracks feed-forward eigenspectrum dynamics across depth. These methods operate on the *depth* axis or on *output* embeddings; ours analyzes the *temporal* axis of latent activations across generation steps, learning nonlinear observables rather than relying on hand-crafted bases.

Koopman Operators and Deep Koopman Autoencoders. Koopman operator theory [10, 14] lifts nonlinear dynamics into a linear observable space, with recent applications to control [12], dynamical similarity [23], and training-time diagnostics [22]. Deep Koopman approaches [13, 2] learn neural observable dictionaries that bypass algebraic factorization; we adapt this architecture to LLM residual stream trajectories.

Sparse Autoencoders for Interpretability. Sparse autoencoders (SAEs) learn overcomplete dictionaries of interpretable features from static activations [17]. Our interpretability pipeline draws direct methodological parallels—max-activating examples, feature labeling, probing, and causal ablation—but operates on *temporal eigenmodes* of a dynamical decomposition, recovering features defined by oscillation frequency and decay rate rather than activation pattern at a single position.

7 Discussion

What KAE Eigenmodes Represent. The learned Koopman operator K represents the average transition structure of the residual stream under a given generation context. It serves as a reference model for regime comparison: individual trajectories can be analyzed by how they deviate from K. The eigenfunctions of K are specific scalar functions of state that may correlate with interpretable generation features, analogous to SAE features but operating on temporal dynamics rather than static activations.

Structural Dominance and Reasoning as Perturbation. Our central finding is that LLM generation dynamics decompose into a dominant structural skeleton (formatting, problem boundaries, turn-taking) and sparse, low-energy transient perturbations where reasoning-specific computation occurs. This echoes findings in neuroscience where cognitive signals are often low-dimensional perturbations on top of dominant motor or sensory dynamics. From an interpretability perspective, this suggests that *reasoning happens in the spectral residuals*—the transient modes that fire briefly during computational transitions.

Revisiting the Eigenspectra Distinction. Applying KAE to the original truthful/untruthful prompting regime of Anonymous [21] yields an important negative result: at $r = 64$, the KAE produces *indistinguishable* eigenspectra for both regimes, in contrast to the clear spectral separation found by EDMD. This suggests that the informational bottleneck of the low-dimensional latent space prioritizes raw reconstruction over the specific phase geometry required to isolate the oscillatory modes discovered by EDMD. The original EDMD approach, operating in the full $D = 25{,}600$-dimensional coordinate basis, preserves fine spectral structure that the KAE's compression discards. This finding highlights a fundamental tradeoff in Koopman approximation: EDMD retains full spectral resolution but scales cubically; KAE scales linearly but may compress away regime-specific spectral signatures.

Limitations. Several limitations warrant discussion. First, each KAE learns its own nonlinear observable basis, making cross-dataset operator comparison ill-posed without a shared encoder or coordinate-free invariants. Second, the probing results show that the eigenbasis does not concentrate information beyond the raw latent, suggesting the current decomposition is faithful but not yet optimally factored for interpretability. Third, full causal verification (Tier 3 logit-shift injection) remains incomplete. Fourth, the spectral radius slightly exceeding unity ($\rho \approx 1.00$–1.01) indicates that the learned system is not strictly contractive, which may complicate long-horizon analysis.

8 Conclusion

We have extended the Latent Spectral Dynamics framework by replacing Hankel-EDMD with Koopman Autoencoders, enabling scalable spectral analysis of LLM generation dynamics. Applied across three datasets and three model families, KAEs reveal an invariant eigenspectrum of autoregressive generation, a resolution scaling phenomenon that progressively uncovers finer dynamical structure, and a structural–transient mode hierarchy in which reasoning-specific signals reside in low-energy transient eigenmodes. These results establish that autoregressive LLM generation admits a principled spectral decomposition, providing a dynamical-systems foundation for temporal interpretability that complements existing static analysis methods.

Key directions for future work include: (i) shared-encoder architectures that enable direct cross-regime spectral comparison; (ii) spectral regularization to enforce contractivity ($\rho(K) \leq 1$); (iii) complete causal verification through logit-shift injection; (iv) scaling to larger models (32B+) and longer generation contexts; and (v) extending the framework to diagnose strategic deception, sycophancy, and other alignment-relevant behaviors as geometrically distinct spectral signatures.

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A KAE Training Details

All KAEs are trained with AdamW (learning rate 10^{-4}, weight decay 10^{-5}, $\beta_1 = 0.9$, $\beta_2 = 0.999$) for 200 epochs with early stopping on validation total loss (patience 20). The loss coefficients are $\alpha = 1.0$ (linear dynamics) and $\beta = 0.5$ (multi-step rollout, uniform weighting across horizons 1 through N). PCA pre-compression retains 512 components, capturing >95% of the variance in the training activations. Data augmentation consists of random trajectory subsampling during training.

B Eigenmode Taxonomy Details

Mode classification thresholds: backbone modes require $|\lambda_j| > 0.95$, $|\theta_j| < 0.05$ rad, and $E_j > 0.01$; oscillatory modes require $|\lambda_j| > 0.80$, $|\theta_j| > 0.05$ rad, and the existence of a conjugate partner within $\Delta\theta < 0.01$ rad; all remaining modes are classified as transient. These thresholds are applied uniformly across all datasets and models.

C Probing Details

Linear probes are logistic regression (classification) or ridge regression (regression) with L_2 regularization $C = 1.0$, trained on 80% of tokens from training trajectories and evaluated on tokens from held-out test trajectories. Reasoning phase labels are derived from chain-of-thought formatting heuristics (problem statement, computation, intermediate result, final answer). Arithmetic context labels are derived from token identity (digits, operators, equals signs within computation spans). Correctness labels are from ground-truth answer comparison. Distance-to-answer is the normalized token count remaining until the final answer token.

D Additional Eigenspectra

This appendix collects supplementary eigenspectrum visualizations supporting the invariant-spectrum, resolution-scaling, and structural–transient taxonomy claims in the main text. Across datasets and model families, the spectra consistently exhibit a near-unity real backbone, scattered oscillatory modes, and a dense lower-energy transient population.

D.1 Eigenmode Taxonomy Details

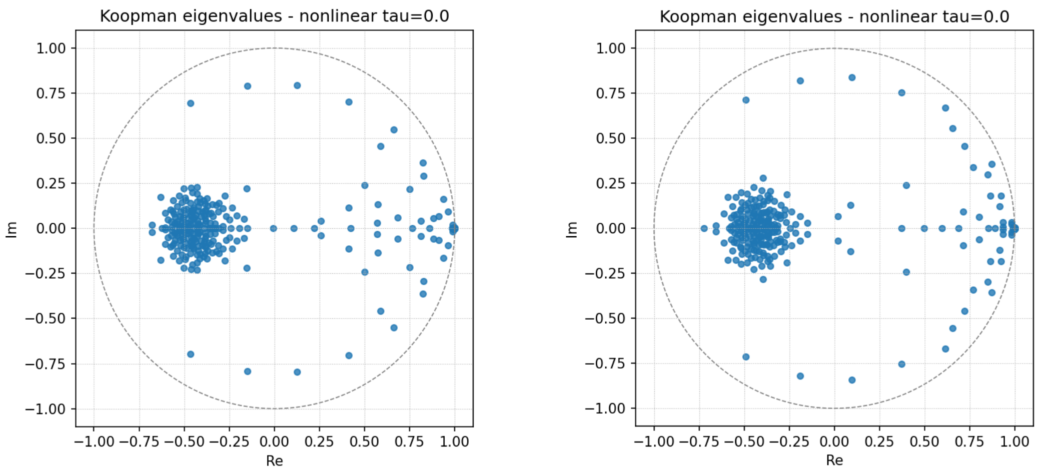
For each mode j, we record the eigenvalue magnitude $|\lambda_j|$, phase $\theta_j = \arg(\lambda_j)$, period $2\pi/|\theta_j|$ when $\theta_j \neq 0$, and empirical energy fraction E_j. The unnormalized energy is computed by taking, for each trajectory, the variance over Hankel positions of $|c_j(k)|^2$, averaging this variance across trajectories, and normalizing across modes so that $\sum_j E_j = 1$.

Let $q_{0.05}$ and $q_{0.50}$ denote the 5th percentile and median of $\{E_j\}$. Modes are assigned in order as follows:

- **Transient modes**: $|\lambda_j| < 0.5$ or $E_j \leq q_{0.05}$.
- **Backbone modes**: remaining modes with $|\lambda_j| \geq 0.9$, near-real phase $|\theta_j| \leq 0.05$, and $E_j \geq q_{0.50}$.
- **Oscillatory modes**: all remaining modes.

D.2 Cross-Architecture GSM8K Spectra

Figure 3 shows GSM8K eigenspectra at $r = 256$ for two additional 14B models. Both exhibit the same broad geometry as Qwen2.5: near-unity real modes, scattered complex-conjugate modes, and a substantial interior population.

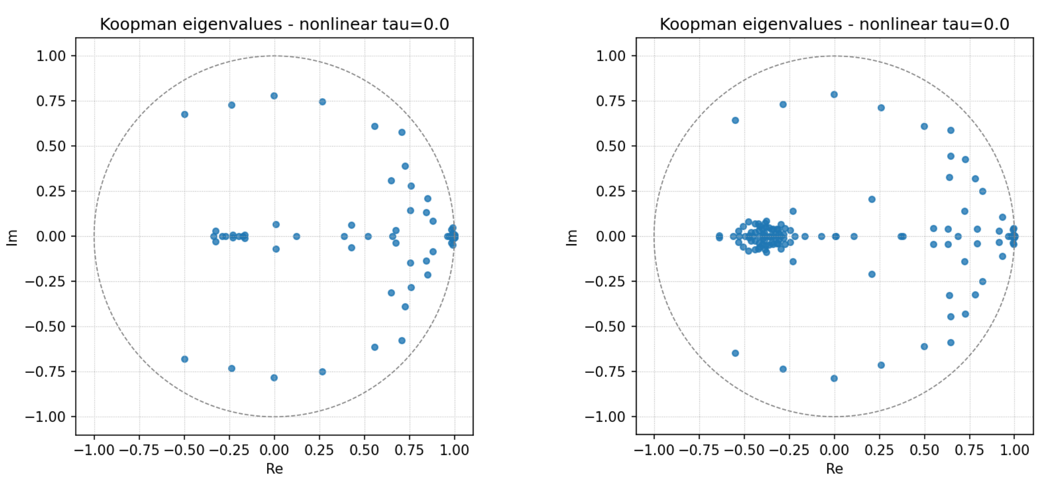


(a) DeepSeek-R1-Distill-Qwen-14B (b) Mistral-3-14B

Figure 3: Additional GSM8K eigenspectra at $r = 256$, $W = 8$ across model families.

D.3 MATH500 Resolution Scaling

Figure 4 shows MATH500 spectra at lower latent dimensions. The near-unity backbone is already visible at small r, while additional interior and oscillatory structure appears as the latent dimension increases.



(a) $r = 64$ (b) $r = 128$

Figure 4: Resolution scaling examples on MATH500 (Qwen2.5-14B-Instruct, $W = 8$).

D.4 High-Resolution WikiText-103 Spectrum

Figure 5 shows the WikiText-103 spectrum at $r = 512$. The higher-dimensional operator reveals a denser interior distribution while retaining the near-unity backbone.

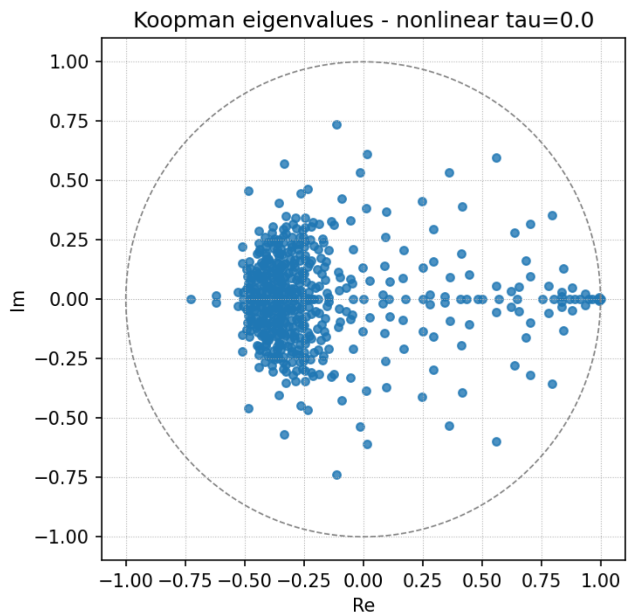
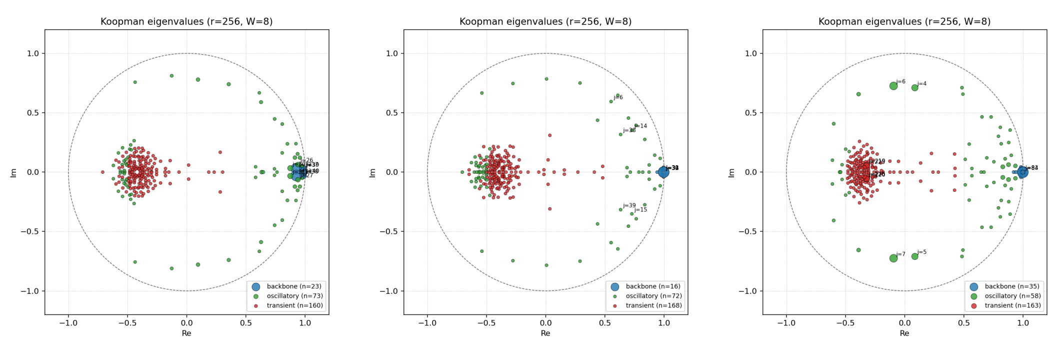


Figure 5: WikiText-103 eigenspectrum at $r = 512$, $W = 8$ (Qwen2.5-14B-Instruct).

D.5 Taxonomized Spectra Across Datasets

Figure 6 visualizes the backbone, oscillatory, and transient partition used in Table 2. The three datasets differ in mode counts and density, but preserve the same structural organization.



(a) GSM8K (b) MATH500 (c) WikiText-103

Figure 6: Taxonomized Koopman eigenspectra at $r = 256$, $W = 8$ (Qwen2.5-14B-Instruct). Blue points denote backbone modes, green points oscillatory modes, and red points transient modes.

E Max-Activation and Feature-Labeling Details

For each mode j, we pool $|c_j(k)|$ over all trajectories and define $\tau_j = \mu_j + 1.5\sigma_j$, where μ_j and σ_j are the empirical mean and standard deviation of $|c_j(k)|$. We extract contiguous runs of at least five Hankel positions satisfying $|c_j(k)| > \tau_j$, rank runs by peak activation, and retain the top 10 text-aligned segments per mode.

The number of candidate runs varies substantially across modes, from 82 to 1470 with median 271. Max-activation windows are treated as evidence for interpretation rather than labels themselves. Across the GSM8K artifacts, high-activation segments concentrate around procedural events such

Table 4: GSM8K max-activation and labeling summary for the $r = 256$, $W = 8$ KAE.

Quantity	Value
Modes scanned	256
Stored segments	2560
Stored segments per mode	10
Candidate runs per mode	82–1470
Median candidate runs per mode	271
Labeled modes	25
Examples shown per labeled mode	6
Procedural / mixed / semantic labels	13 / 10 / 2
High / medium confidence labels	13 / 12

as problem-to-solution onset, equation setup, arithmetic execution, answer emission, follow-up-question handoff, and prompt reset.

Table 5: Representative GSM8K modes from the $r = 256$, $W = 8$ KAE interpretability pipeline.

Mode(s)	**Class**	**Label Type**	**Max-Activation Pattern**
90	Transient	Procedural	Follow-up-question boundary before a new computation begins; strongest windows occur near handoff markers such as empty brackets or answer-boundary delimiters.
32/33	Backbone pair	Procedural	Final-answer production followed by transition to the next prompt; windows peak around boxed/bracketed numeric answers and subsequent “Human:” prompt onset.
47/48	Backbone pair	Procedural / mixed	Problem-statement-to-solution boundary; windows peak near the end of the word problem and the beginning of structured solution text such as “To find...”.
51/52	Backbone pair	Mixed	Question-boundary and solution-onset detection; activations cluster around delimiter tokens marking the transition from problem intake to chain-of-thought generation.
10/11	Control pair	Semantic	Surface-template-associated semantic controls; activations are tied to repeated text patterns such as “be denoted” rather than broad conceptual content.